Devices and peripherals 2.0

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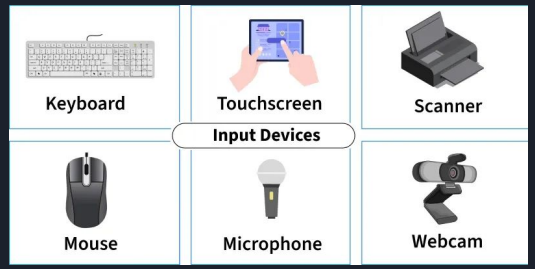
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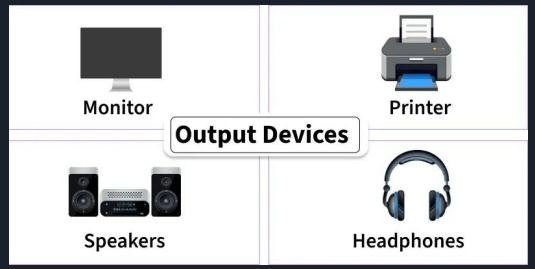
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1.Types of peripherals

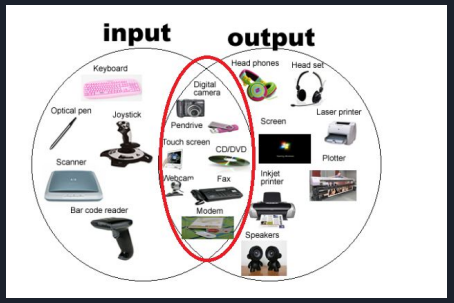
1. Input devices



2. Output devices



2. Input/Output devices



2. Examples of input devices

Keyboard



Mouse



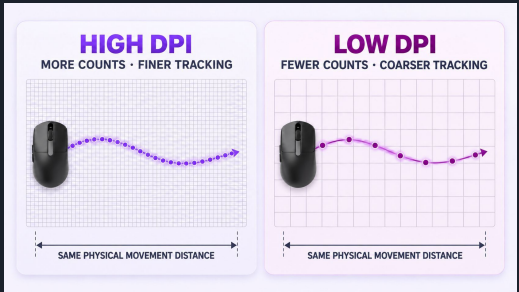
Controller



2. 1 Specifying on the characteristics of an input device

Mouse:

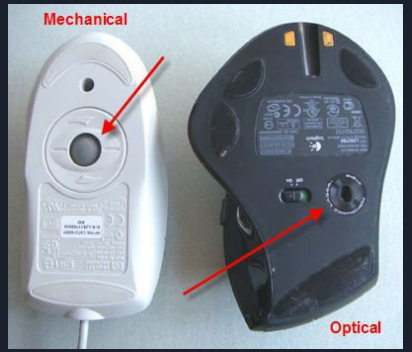
Sensor & DPI Resolution: Determines tracking accuracy (optical vs. laser) and sensitivity (up to 25,000+ DPI).



Polling Rate (Hz): The frequency at which the mouse reports to the CPU, defining input latency .



Switch Type (Optical vs. Mechanical): Optical switches use light beams to register clicks instantly, eliminating mechanical debounce delay.



3. Examples of output devices

Monitor



Headphones



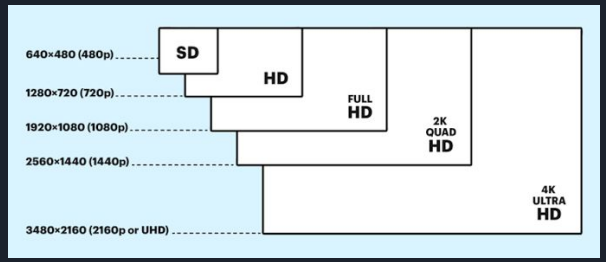
Speakers



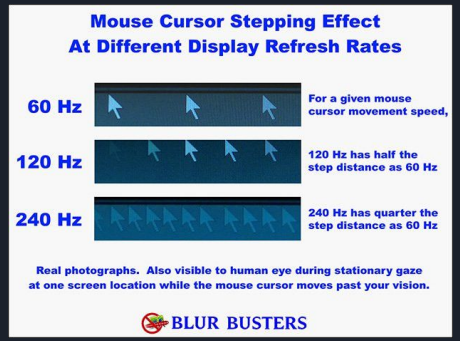
3. 1 Specifying on the characteristics of an output device

Monitor:

Resolution & Panel Type (IPS, VA, OLED): Determines image sharpness and color accuracy. Panel technology dictates contrast and viewing angles, while resolution (e.g., 1080p, 4K) sets the pixel density.



Refresh Rate (Hz): The frequency at which the screen updates the image per second, defining visual smoothness and input responsiveness (e.g., 60Hz for office work, 120Hz+ for gaming).



Response Time (ms): The time it takes for a pixel to transition from one color to another (Gray-to-Gray). Lower times (1ms) prevent ghosting and motion blur in fast-paced scenes.



4. Examples of input/output devices

Touchscreen



Printer with scanner



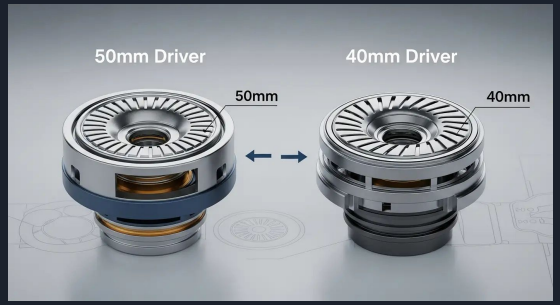
Gaming Headsets



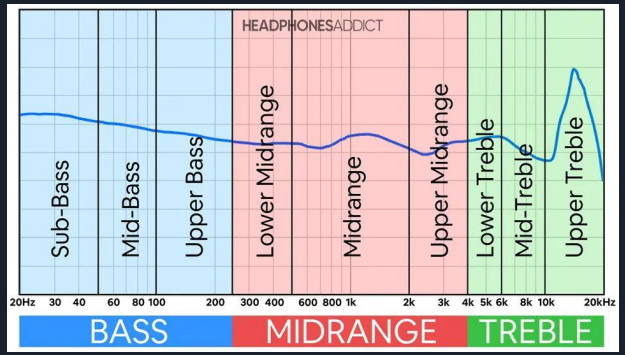
4. 1 Specifying on the characteristics of an input and output device

Gaming headsets:

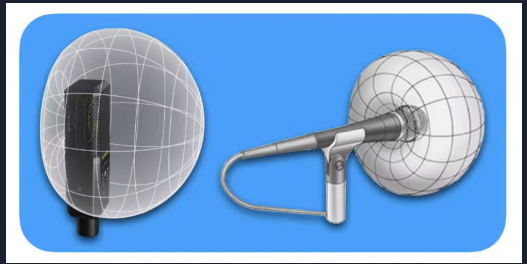
Driver Size (Output): Usually 40mm or 50mm neodymium magnets. Larger drivers push more air, producing deeper bass and clearer audio for games.



Frequency Response (Output): Indicates the range of sounds the headset can reproduce. The standard is 20 Hz (deep bass) to 20,000 Hz (high treble), which covers the entire human hearing spectrum.



Microphone Pattern (Input): Most use a "Unidirectional" (or Cardioid) pattern. This means the microphone only captures sound from the front (your voice) and blocks background noise like keyboard clicks.



5.Differences between devices that require drivers and those that have plug and play

Plug and Play Devices

Instant connection: They function immediately when connected. The Operating System (OS) already includes generic, built-in software for them.

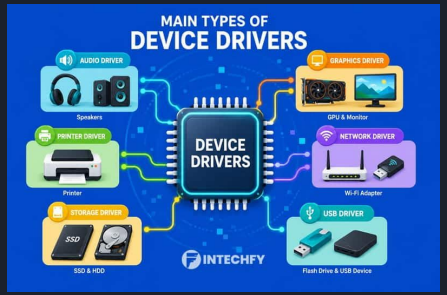
Standard protocols: They use universal communication rules, such as USB-HID (Human Interface Device) for standard mice and keyboards, allowing the PC to recognize them automatically.

Devices Requiring Specific Drivers

Custom software needed: They require specialized manufacturer software installed on the computer to function correctly or to unlock all their features.

Advanced translation: The specific driver acts as a translator for complex, non-standard instructions. It enables proprietary features like custom macro keys, RGB lighting control, or precise DPI tuning.





6. Main ports/wires to connect external devices

USB (Universal Serial Bus) : The universal standard for connecting most peripherals (mouse, keyboards, printers).It transmits both data and electrical power.



HDMI & DisplayPort:
High-speed digital ports used for video and audio output to monitors. DisplayPort is usually preferred for PC gaming as it offers higher bandwidth for faster refresh rates and multiple monitor setups, while HDMI is the standard for TVs and consoles.



Ethernet (RJ45): The main port for wired network connectivity. It provides a more stable, lower-latency connection than Wi-Fi, supporting Gigabit (1000 Mbps) or even faster data transfer speeds.



6. 1 Specifying on the characteristics of the most used ports/wires

USB :

USB 2.0 (Legacy) : Up to 480 Mbps. Used for low-speed peripherals like keyboards and basic mouse.



USB 3.0 / 3.1 / 3.2 (SuperSpeed): 5 Gbps to 20 Gbps (over 10x faster). Used for external hard drives and fast flash drives.



USB4 (Ultra-Fast) : 40 Gbps to 80 Gbps. Exclusive to USB-C ports, capable of transmitting high-speed data and 4K video simultaneously.

